

TIC 334422203

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 2.491163 d (ALIASED, see below)
-> corrected period: 4.982326 d
Epoch (t0) 1598.3133 BTJD
Duration 0.0840 d
Depth 9.583 % (95,828 ppm)
SNR 180.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

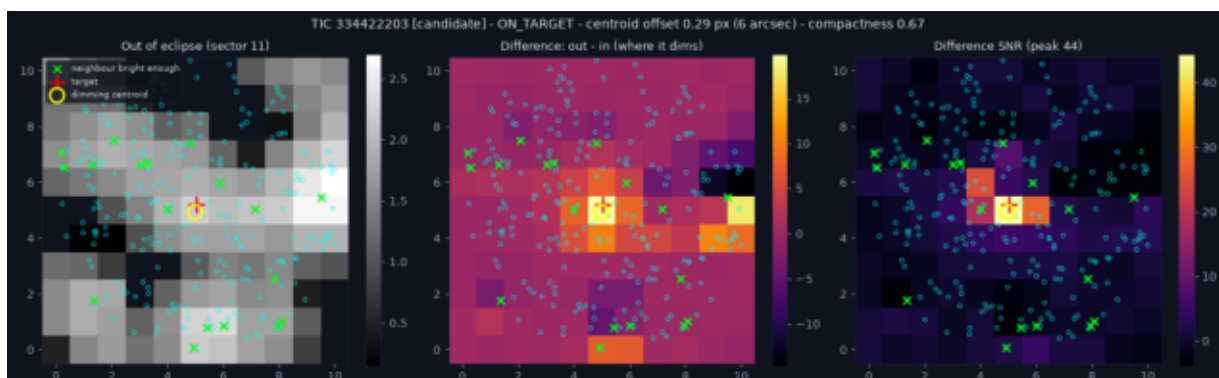
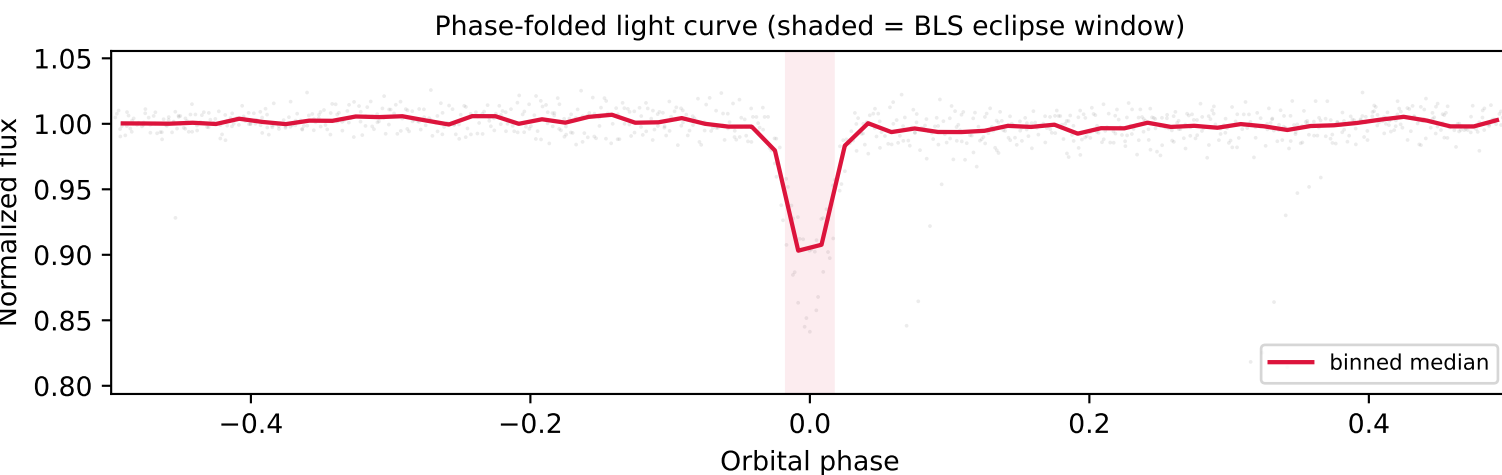
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 7 (gate ≥ 3)
Robust depth ratio 0.96 (gate ≥ 0.50)
Robust depth sigma 43.3 (gate ≥ 7.0)
Odd/even z-score 3.74 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 11 light-curve dip in this sector: 32.6 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 44.36 (gate ≥ 5.0) Compactness 0.67 (gate ≥ 0.50)
Dimming centroid offset 0.29 px (6 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 334422206
0.99 px away, dTmag +1.90 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.74 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 4.982326 d, in any write-up.